

Industrial Practical Training - I

on

Full Stack Web Development

submitted in partial fulfillment of the requirements

for the award of the degree

of

Bachelor of Technology

in

Computer Science Engineering

By

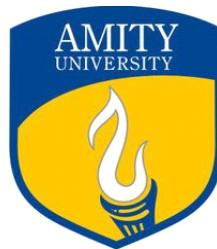
Rajat Dua

Enrollment No. A60205223023

Under the guidance of

Dr. Ashok Kumar Shrivastava

Associate Professor



**Department of Computer Science Engineering
Amity School of Engineering & Technology
Amity University Madhya Pradesh, Gwalior
December 2025**



**Department of Computer Science Engineering
Amity School of Engineering and Technology
Amity University Madhya Pradesh, Gwalior**

DECLARATION

I, **Rajat Dua**, student of Bachelor of Technology in Computer Science Engineering hereby declare that the practical training report entitled “**FULL STACK WEB DEVELOPMENT**” which is submitted by me to Department of Computer Science Engineering, Amity School of Engineering & Technology, Amity University Madhya Pradesh, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

Rajat Dua

Date:

(Enrollment No. – A60205223023)



**Department of Computer Science Engineering
Amity School of Engineering and Technology
Amity University Madhya Pradesh, Gwalior**

CERTIFICATE

This is to certify that **Rajat Dua (Enrollment NO. A60205223023)**, student of B.Tech(C.S.E) V semester, Department of Computer Science Engineering, ASET, Amity University Madhya Pradesh, has done his practical training entitled “**FULL STACK WEB DEVELOPMENTS**” under my guidance and supervision during “**16 July 2025 – 16 August 2025**”.

The work was satisfactory. He has shown complete dedication and devotion to the given project work.

Date:

(Dr. Ashok Kumar Shrivastava)

Associate Professor

Supervisor

External Examiner

(Prof. Dr. Vikas Thada)

Head of the Department



CERTIFICATE OF INTERNSHIP

This certificate awarded to
Rajat Dua

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC Approved
internship program as

Full Stack Developer Intern

Conducted from 16/07/2025 to 16/08/2025



AICTE ID - CORPORATE6759d549ce59e1733940553

Dr. Anurag Shrivastava
Program Manager

Letter of Recommendation



Company NCS ID - E19E86-011658828892

Date of Issue - 16/08/2025

To Whom It May Concern, I am writing this letter to highly recommend
Rajat Dua

for their exceptional performance and unwavering dedication during their tenure as an intern at Codec Technologies. Their time with us has been marked by an exemplary work ethic, a profound passion for the field, and an impressive ability to excel in their assigned tasks. Throughout their internship, they consistently demonstrated a high level of professionalism and initiative, often exceeding our expectations. They possess remarkable technical skills and a strong understanding of the industry, which they applied effectively to tackle complex projects. Their enthusiasm for their work was evident, as they displayed outstanding problem-solving abilities and a keen aptitude for learning.

Based on their outstanding performance and evident potential, I have no doubt that they will continue to thrive and make significant contributions in their future endeavors. It is with great pleasure and confidence that I recommend them for any future opportunities or positions they may seek. Their dedication and capabilities will undoubtedly be a valuable asset to any organization.

Sincerely,

Dr. Anurag Shrivastava
Program Manager



ACKNOWLEDGEMENT

We are very much thankful to our Honourable Pro Chancellor **Lt Gen. V. K. Sharma AVSM (Retd)** for allowing us to carry out our project. We would also like to thank **Prof.(Dr.) R. S. Tomar**, Honourable Vice Chancellor, Amity University Madhya Pradesh for his valuable support. We would also like to thank Respected **Prof. (Dr.) M. P. Kaushik**, Pro-Vice Chancellor (Research), Amity University Madhya Pradesh for his guidance.

I extend my sincere thanks to **Prof. (Dr.) Vikas Thada**, HOI, Amity School of Engineering and Technology, Amity University Madhya Pradesh, Gwalior for his guidance and support for the selection of appropriate industry for my practical training. I would also like to thank **Prof. (Dr.) Vikas Thada** Head of Department (CSE), for his kind concern throughout the practical training.

I am also very grateful to **Dr. Ashok Kumar Shrivastava**, Associate Professor, Department of Computer Science, Amity School of Engineering and Technology, Amity University Madhya Pradesh and **Dr. Anurag Shrivastava**, CODEC TECHNOLOGIES . My internal and external supervisor respectively, for their constant guidance and encouragement provided in this endeavour.

I am also thankful to the whole staff of CODEC TECHNOLOGIES, for the co-operation and giving me friendly environment which made very comfortable for learning and that of ASET, AUMP for teaching and helping me always. Last but not the least I would like to thank my parents and friends for their constant support.

Rajat Dua

(Enrollment No. – A60205223023)

ABSTRACT

This internship report presents a detailed account of the work completed during a four-week training program in full-stack web development at CODEC TECHNOLOGIES.

The focus of the project was the creation of a real-time chat application that allows multiple users to communicate instantly within a shared digital environment. To achieve this, modern web technologies were combined. Node.js and Express.js formed the server layer, while Socket.IO enabled real-time, bidirectional communication. The client interface was designed using HTML, CSS, and JavaScript to ensure accessibility, responsiveness, and a user-friendly experience.

The report outlines the motivation behind selecting a real-time communication tool, followed by an explanation of the application's core architecture, the development approaches adopted, and the insights gained during each stage. Emphasis is placed on understanding how WebSocket-based applications operate, how user events are synchronized across clients, and how a simple interface can be expanded into more advanced, production-ready systems. The internship provided extensive exposure to practical full-stack development, improving technical confidence and offering first-hand experience in creating interactive web solutions.

Keywords: Full-Stack Web Development, Real-Time Chat Application, Node.js, Express.js, Socket.IO, WebSocket Communication, Bidirectional Messaging, Instant Communication, Multi-User Interaction, Frontend Development, HTML, CSS, Javascript, Responsive Interface, User-Friendly UI, Application Architecture, Event Synchronization, Real-Time Systems, Development Methodology, Practical Experience, Interactive Web Solutions.

LIST OF FIGURES

Figure No.	Figure Caption	Page No.
Figure 5.1	User Login and Chat Room Access Screen	13
Figure 5.2	Active Users Display Panel	13
Figure 5.3	Main Chat Room Interface	14
Figure 5.4	Multi-User Message Interaction View	15

LIST OF TABLES

Table No.	Table Caption	Page No.
Table 1.1	Technologies and Tools Utilized	2
Table 3.1	WebSocket / Socket.IO Events	9
Table 4.1	Testing and Verification Checklist	11

LIST OF ABBREVIATIONS

S.No.	Abbreviations	Full Form
1.	API	Application Programming Interface
2.	UI	User Interface
3.	UX	User Experience
4.	DOM	Document Object Model
5.	WS	WebSocket
6.	HTTP	HyperText Transfer Protocol
7.	DB	Database
8.	IDE	Integrated Development Environment
9.	JS	JavaScript
10.	CID	Client Identifier

CONTENTS

Content	Page No.
Front Page	i
Declaration by Student	ii
Certificate by Supervisor (Forwarded by HOD/HOI)	iii
Certificate by Company	iv
Letter of Recommendation	v
Acknowledgement	vi
Abstract	vii
List of Figures	viii
List of Tables	ix
List of Abbreviations	x
Chapter 1. Introduction	1-3
Chapter 2. Modular Description Of Job	4-6
Chapter 3. Methodology	7-9
Chapter 4. Analysis	10-11
Chapter 5. Graphical User Interface	12-15
Chapter 6. Conclusion and Future Prospects	16-17
Chapter 7. Summary	18
Bibliography	19